

Mathematics for Data Science

Reference Proof Sheet – Lecture 2

Matrices, Linear Systems, Invertibility and Change of Basis

M1 – AI / Data Science

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1 Matrix multiplication

1.1 Associativity, distributivity and identity matrices

Proposition – Basic properties of matrix multiplication

Let

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{n \times p}, \quad C \in \mathbb{R}^{p \times q}.$$

Then

$$(AB)C = A(BC).$$

Moreover, whenever the matrix sizes are compatible,

$$A(B + C) = AB + AC, \quad (A + B)C = AC + BC.$$

Finally, for every $A \in \mathbb{R}^{m \times n}$,

$$I_m A = A, \quad A I_n = A.$$

Proof. We prove the identities entry by entry.

Associativity. For $1 \leq i \leq m$ and $1 \leq j \leq q$,

$$\begin{aligned} ((AB)C)_{ij} &= \sum_{\ell=1}^p (AB)_{i\ell} c_{\ell j} \\ &= \sum_{\ell=1}^p \left(\sum_{k=1}^n a_{ik} b_{k\ell} \right) c_{\ell j} \\ &= \sum_{k=1}^n a_{ik} \left(\sum_{\ell=1}^p b_{k\ell} c_{\ell j} \right) \\ &= \sum_{k=1}^n a_{ik} (BC)_{kj} \\ &= (A(BC))_{ij}. \end{aligned}$$

The two matrices have the same entries, hence $(AB)C = A(BC)$.

Distributivity. For example,

$$\begin{aligned} (A(B + C))_{ij} &= \sum_{k=1}^n a_{ik} (b_{kj} + c_{kj}) \\ &= \sum_{k=1}^n a_{ik} b_{kj} + \sum_{k=1}^n a_{ik} c_{kj} \\ &= (AB)_{ij} + (AC)_{ij}. \end{aligned}$$

Therefore $A(B + C) = AB + AC$. The proof of right distributivity is identical.

Identity matrices. Recall that $(I_m)_{ik}$ equals 1 when $i = k$ and 0 otherwise. Hence

$$(I_m A)_{ij} = \sum_{k=1}^m (I_m)_{ik} a_{kj} = a_{ij}.$$

Thus $I_m A = A$. Similarly,

$$(A I_n)_{ij} = \sum_{k=1}^n a_{ik} (I_n)_{kj} = a_{ij},$$

so $AI_n = A$. □

Remark – Matrix multiplication is not commutative
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In general, $AB \neq BA$, even when both products are defined.

For example,

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

give

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

Thus $AB \neq BA$.

2 Matrices and linear maps

2.1 A matrix defines a linear map

Proposition

Let $A \in \mathbb{R}^{m \times n}$. The map

$$T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad x \mapsto Ax,$$

is linear.

Proof. Let $x, y \in \mathbb{R}^n$ and let $\alpha, \beta \in \mathbb{R}$. By distributivity of matrix multiplication and compatibility with scalar multiplication,

$$\begin{aligned} T_A(\alpha x + \beta y) &= A(\alpha x + \beta y) \\ &= \alpha Ax + \beta Ay \\ &= \alpha T_A(x) + \beta T_A(y). \end{aligned}$$

Hence T_A is linear. □

2.2 Matrix representation of a linear map

Proposition – Matrix of a linear map

Let V and W be finite-dimensional vector spaces with bases
--

$$\mathcal{B} = (v_1, \dots, v_n), \quad \mathcal{C} = (w_1, \dots, w_m),$$

and let $L : V \rightarrow W$ be linear. For every $j \in \{1, \dots, n\}$, there exist unique scalars $a_{1j}, \dots, a_{mj}$ such that

$$L(v_j) = \sum_{i=1}^m a_{ij} w_i.$$

If $A = (a_{ij}) \in \mathbb{R}^{m \times n}$, then for every $x \in V$,
--

$$[L(x)]_{\mathcal{C}} = A[x]_{\mathcal{B}}.$$

Thus A is the matrix representation of L from $\mathcal{B}$ to $\mathcal{C}$.
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Proof.

Existence and uniqueness of the columns. For each basis vector v_j , the vector $L(v_j)$ belongs to W . Since $\mathcal{C} = (w_1, \dots, w_m)$ is a basis of W , every vector of W has unique coordinates in this basis. Therefore there exist unique scalars $a_{1j}, \dots, a_{mj}$ such that

$$L(v_j) = a_{1j}w_1 + \dots + a_{mj}w_m.$$

These coordinates form the j th column of A .

Action on an arbitrary vector. Let

$$[x]_{\mathcal{B}} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \text{so that} \quad x = \sum_{j=1}^n x_j v_j.$$

By linearity,

$$\begin{aligned} L(x) &= \sum_{j=1}^n x_j L(v_j) \\ &= \sum_{j=1}^n x_j \sum_{i=1}^m a_{ij} w_i \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} x_j \right) w_i. \end{aligned}$$

Hence the i th coordinate of $L(x)$ in the basis $\mathcal{C}$ is $\sum_{j=1}^n a_{ij} x_j$, which is precisely the i th entry of $A[x]_{\mathcal{B}}$. Therefore

$$[L(x)]_{\mathcal{C}} = A[x]_{\mathcal{B}}.$$

□

2.3 Matrix multiplication and composition

Proposition – Matrix of a composition

Let U, V, W be finite-dimensional vector spaces with bases $\mathcal{A}, \mathcal{B}, \mathcal{C}$. Let $L : U \rightarrow V$ and $M : V \rightarrow W$ be linear maps. Then

$$[M \circ L]_{\mathcal{C} \leftarrow \mathcal{A}} = [M]_{\mathcal{C} \leftarrow \mathcal{B}} [L]_{\mathcal{B} \leftarrow \mathcal{A}}.$$

Proof. Let

$$A = [L]_{\mathcal{B} \leftarrow \mathcal{A}}, \quad B = [M]_{\mathcal{C} \leftarrow \mathcal{B}}.$$

For every $x \in U$,

$$\begin{aligned} [(M \circ L)(x)]_{\mathcal{C}} &= B[L(x)]_{\mathcal{B}} \\ &= BA[x]_{\mathcal{A}}. \end{aligned}$$

Therefore the matrix of $M \circ L$ is BA . □

3 Range, kernel and rank of a matrix

3.1 The range is the column space

Proposition

Let $A \in \mathbb{R}^{m \times n}$, with columns $a_1, \dots, a_n \in \mathbb{R}^m$. Then

$$\text{range}(A) = \text{Col}(A) = \text{span}(a_1, \dots, a_n).$$

Proof. Let

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n.$$

By the definition of matrix–vector multiplication,

$$Ax = x_1 a_1 + \dots + x_n a_n.$$

Thus every vector of the form Ax is a linear combination of the columns of A , so

$$\text{range}(A) \subseteq \text{span}(a_1, \dots, a_n).$$

Conversely, every linear combination $x_1 a_1 + \dots + x_n a_n$ is equal to Ax for $x = (x_1, \dots, x_n)^\top$. Hence

$$\text{span}(a_1, \dots, a_n) \subseteq \text{range}(A).$$

Therefore the two sets are equal. □

3.2 Kernel and injectivity

Proposition – Kernel and injectivity

Let $A \in \mathbb{R}^{m \times n}$. The map $x \mapsto Ax$ is injective if and only if

$$\ker(A) = \{0\}.$$

Proof. This is the kernel criterion for linear maps from Lecture 1, applied to the linear map $T_A : x \mapsto Ax$. For completeness, we give the short proof.

Suppose first that T_A is injective. If $x \in \ker(A)$, then $Ax = 0 = A0$. Injectivity implies $x = 0$. Thus $\ker(A) = \{0\}$.

Conversely, suppose $\ker(A) = \{0\}$ and assume $Ax = Ay$. Then

$$A(x - y) = 0,$$

so $x - y \in \ker(A)$. Hence $x - y = 0$, and therefore $x = y$. Thus T_A is injective. □

3.3 Rank–nullity for matrices

Theorem – Rank–nullity for matrices

Let $A \in \mathbb{R}^{m \times n}$. Then

$$n = \text{rank}(A) + \dim(\ker(A)).$$

Proof. Consider the linear map

$$T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad x \mapsto Ax.$$

Its kernel is exactly $\ker(A)$, and its range is the column space of A . Therefore

$$\text{rank}(T_A) = \dim(\text{range}(T_A)) = \text{rank}(A).$$

Applying the rank–nullity theorem from Lecture 1 to T_A gives

$$\dim(\mathbb{R}^n) = \text{rank}(T_A) + \dim(\ker(T_A)).$$

Since $\dim(\mathbb{R}^n) = n$, the result follows. $\square$

3.4 The square case

Proposition – Equivalent statements for a square matrix
Let $A \in \mathbb{R}^{n \times n}$. The following statements are equivalent: (1) the map $x \mapsto Ax$ is bijective; (2) $\text{range}(A) = \mathbb{R}^n$; (3) $\ker(A) = \{0\}$; (4) $\text{rank}(A) = n$.

Let $A \in \mathbb{R}^{n \times n}$. The following statements are equivalent:

- (1) the map $x \mapsto Ax$ is bijective;
- (2) $\text{range}(A) = \mathbb{R}^n$;
- (3) $\ker(A) = \{0\}$;
- (4) $\text{rank}(A) = n$.

Proof. By definition, the map $x \mapsto Ax$ is surjective if and only if $\text{range}(A) = \mathbb{R}^n$. Since $\text{range}(A)$ is a subspace of $\mathbb{R}^n$, this is equivalent to

$$\dim(\text{range}(A)) = n,$$

i.e. to $\text{rank}(A) = n$.

By rank–nullity,

$$n = \text{rank}(A) + \dim(\ker(A)).$$

Therefore $\text{rank}(A) = n$ if and only if $\dim(\ker(A)) = 0$, which is equivalent to $\ker(A) = \{0\}$. By the kernel criterion, this is equivalent to injectivity.

Thus, for a linear map from $\mathbb{R}^n$ to $\mathbb{R}^n$, injectivity and surjectivity are equivalent. Either one is therefore equivalent to bijectivity, proving the four-way equivalence. $\square$

4 Linear systems and Gaussian elimination

4.1 Structure of the solution set

Proposition – General solution of a consistent linear system
Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Assume that x_p is one particular solution of $Ax = b$. Then the set of all solutions is $x_p + \ker(A) = \{x_p + z \mid z \in \ker(A)\}.$

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Assume that x_p is one particular solution of $Ax = b$. Then the set of all solutions is

$$x_p + \ker(A) = \{x_p + z \mid z \in \ker(A)\}.$$

Proof. Let x be any solution of $Ax = b$. Since $Ax_p = b$ as well,

$$A(x - x_p) = Ax - Ax_p = b - b = 0.$$

Hence $x - x_p \in \ker(A)$, so $x = x_p + z$ for some $z \in \ker(A)$.

Conversely, let $z \in \ker(A)$. Then

$$A(x_p + z) = Ax_p + Az = b + 0 = b.$$

Thus every vector of the form $x_p + z$ is a solution. Therefore the solution set is exactly $x_p + \ker(A)$.
□

4.2 Elementary row operations preserve the solution set

Proposition

The following elementary transformations of a linear system preserve its solution set:

- (1) exchanging two equations;
- (2) adding a scalar multiple of one equation to another equation;
- (3) multiplying one equation by a nonzero scalar.

Proof. Each operation is reversible by an operation of the same type.

Exchanging two equations does not change which vectors satisfy all equations. If one equation is replaced by itself plus λ times another equation, we recover the original equation by subtracting λ times the unchanged equation. If one equation is multiplied by $\alpha \neq 0$, we recover the original equation by multiplying by $1/\alpha$.

Thus every solution of the original system is a solution of the transformed system, and reversibility gives the converse inclusion. Hence the solution sets are identical. □

4.3 Row space and row equivalence

Lemma – Row operations preserve the row space

If matrices A and R are row-equivalent, then

$$\text{Row}(A) = \text{Row}(R).$$

Proof. It is enough to consider one elementary row operation. After such an operation, every new row is a linear combination of the old rows. Therefore the new row space is contained in the old row space.

Since every elementary row operation is reversible by another elementary row operation, every old row is also a linear combination of the new rows. Hence the reverse inclusion holds. Thus the two row spaces are equal. Applying this argument along a sequence of row operations proves the result for row-equivalent matrices. □

4.4 Uniqueness of reduced row-echelon form

Theorem – Uniqueness of reduced row-echelon form

Every $m \times n$ matrix is row-equivalent to a unique reduced row-echelon form.

Proof.

Existence. Gaussian elimination produces a row-echelon form. By scaling each pivot to 1 and then eliminating all entries above each pivot, one obtains a reduced row-echelon form. Hence every matrix is row-equivalent to at least one reduced row-echelon form.

A property of the row space of an RREF matrix. Let R be in reduced row-echelon form, with nonzero rows $r_1, \dots, r_k$ and pivot columns

$$p_1 < p_2 < \dots < p_k.$$

Every nonzero vector $u \in \text{Row}(R)$ has its first nonzero coordinate in one of the pivot columns.

Indeed, write

$$u = \alpha_1 r_1 + \dots + \alpha_k r_k$$

and let j be the smallest index such that $\alpha_j \neq 0$. All rows $r_{j+1}, \dots, r_k$ have zeros in columns before p_j , and the rows $r_1, \dots, r_{j-1}$ do not occur in the combination by minimality of j . Moreover, the pivot column p_j contains a 1 in row j and zeros in all other nonzero rows. Hence the first nonzero coordinate of u is at column p_j .

It follows that the set of pivot columns of an RREF matrix is determined entirely by its row space.

Uniqueness. Suppose R and S are two reduced row-echelon forms row-equivalent to the same matrix A . By the preceding lemma,

$$\text{Row}(R) = \text{Row}(A) = \text{Row}(S).$$

Hence R and S have the same pivot columns, say $p_1 < \dots < p_k$.

Let r_i and s_i denote their i th nonzero rows. Because both matrices are in reduced row-echelon form, both r_i and s_i have entry 1 in column p_i and entry 0 in every other pivot column. Therefore the difference

$$d_i = r_i - s_i$$

belongs to the common row space and has entry 0 in every pivot column. If $d_i \neq 0$, the property proved above says that its first nonzero coordinate must occur in a pivot column, which is impossible. Thus $d_i = 0$, so $r_i = s_i$ for every i . The zero rows also agree. Hence $R = S$. $\square$

4.5 Rank and pivots

Proposition – Rank from row reduction

Let $A \in \mathbb{R}^{m \times n}$, and let R be any row-echelon form of A . If R has r pivots, then

$$\text{rank}(A) = r.$$

Moreover:

- (1) the pivot columns of the *original matrix* A form a basis of $\text{Col}(A)$;
- (2) the nonzero rows of R form a basis of $\text{Row}(A)$;
- (3) consequently,

$$\dim(\text{Col}(A)) = \dim(\text{Row}(A)).$$

Proof. It is convenient to use a reduced row-echelon form; this does not change the pivot positions. Let R be the RREF of A , with r pivots.

Column relations are preserved by row operations. A sequence of elementary row operations amounts to left-multiplication by an invertible matrix E :

$$R = EA.$$

Let $a_1, \dots, a_n$ be the columns of A and $r_1, \dots, r_n$ the corresponding columns of R . Then $r_j = Ea_j$ for every j . For scalars $\lambda_1, \dots, \lambda_n$,

$$\lambda_1 r_1 + \dots + \lambda_n r_n = E(\lambda_1 a_1 + \dots + \lambda_n a_n).$$

Since E is invertible, the left-hand side is zero if and only if the expression inside E is zero. Therefore A and R have exactly the same linear-dependence relations among corresponding columns.

Pivot columns. In RREF, the r pivot columns are linearly independent: restricted to the nonzero rows, they are the standard coordinate vectors. Every nonpivot column is a linear combination of the pivot columns, with coefficients given by its entries in the pivot rows. Thus the pivot columns of R form a basis of $\text{Col}(R)$. By preservation of column relations, the corresponding columns of the original matrix A are linearly independent and span all columns of A . Hence they form a basis of $\text{Col}(A)$, and

$$\text{rank}(A) = \dim(\text{Col}(A)) = r.$$

Rows. Row operations preserve the row space, so $\text{Row}(A) = \text{Row}(R)$. The nonzero rows of R span $\text{Row}(R)$ by definition. They are linearly independent because their leading entries occur in distinct pivot columns. Hence they form a basis of $\text{Row}(A)$ and

$$\dim(\text{Row}(A)) = r = \dim(\text{Col}(A)).$$

This proves, in particular, that row rank equals column rank. □

4.6 Consistency and number of solutions

Proposition – Rank criterion for a linear system

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Then:

(1) $Ax = b$ has no solution if and only if

$$\text{rank}(A) < \text{rank}([A \mid b]).$$

(2) If

$$\text{rank}(A) = \text{rank}([A \mid b]) = n,$$

then $Ax = b$ has a unique solution.

(3) If

$$\text{rank}(A) = \text{rank}([A \mid b]) < n,$$

then $Ax = b$ has infinitely many solutions.

Proof.

Consistency. The equation $Ax = b$ is solvable exactly when b belongs to the span of the columns of A . Appending b as an additional column therefore does not increase the dimension of the column space exactly when the system is consistent. Thus

$$Ax = b \text{ is consistent} \iff \text{rank}([A \mid b]) = \text{rank}(A).$$

This proves (1).

Unique solution. Assume the system is consistent and $\text{rank}(A) = n$. Then rank-nullity gives

$$\dim(\ker(A)) = n - \text{rank}(A) = 0.$$

Hence $\ker(A) = \{0\}$. If x_p is one solution, the full solution set is $x_p + \ker(A) = \{x_p\}$. Therefore the solution is unique.

Infinitely many solutions. Assume the system is consistent and $\text{rank}(A) < n$. Then

$$\dim(\ker(A)) = n - \text{rank}(A) > 0.$$

Choose $z \in \ker(A)$ with $z \neq 0$, and let x_p be one particular solution. For every $t \in \mathbb{R}$,

$$x_p + tz$$

is a solution. Distinct values of t give distinct vectors because $z \neq 0$. Hence there are infinitely many solutions. $\square$

5 Special matrices

5.1 Identity matrix

Proposition – Properties of the identity matrix

For every $x \in \mathbb{R}^n$ and every $A \in \mathbb{R}^{n \times n}$,

$$I_n x = x, \quad A I_n = I_n A = A.$$

Consequently,

$$I_n I_n = I_n, \quad I_n^{-1} = I_n.$$

Proof. The identities $A I_n = A$ and $I_n A = A$ were proved in Section 1. Taking A to be the $n \times 1$ matrix whose only column is x gives $I_n x = x$. Taking $A = I_n$ gives $I_n I_n = I_n$. Finally, the definition of an inverse requires a matrix B satisfying $I_n B = B I_n = I_n$; choosing $B = I_n$ proves $I_n^{-1} = I_n$. $\square$

5.2 Transpose

Proposition – Properties of the transpose

Let A and B be matrices of compatible sizes and let $\alpha \in \mathbb{R}$. Then

- (i) $(A^\top)^\top = A$;
- (ii) $(A + B)^\top = A^\top + B^\top$;
- (iii) $(\alpha A)^\top = \alpha A^\top$;
- (iv) $(AB)^\top = B^\top A^\top$.

Proof. The first three identities follow directly from the definition $(A^\top)_{ij} = A_{ji}$. For instance,

$$((A + B)^\top)_{ij} = (A + B)_{ji} = a_{ji} + b_{ji} = (A^\top + B^\top)_{ij}.$$

For the product identity, let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. For $1 \leq i \leq p$ and $1 \leq j \leq m$,

$$\begin{aligned} ((AB)^\top)_{ij} &= (AB)_{ji} \\ &= \sum_{k=1}^n a_{jk} b_{ki} \\ &= \sum_{k=1}^n (B^\top)_{ik} (A^\top)_{kj} \\ &= (B^\top A^\top)_{ij}. \end{aligned}$$

Hence $(AB)^\top = B^\top A^\top$. □

5.3 Diagonal matrices

Proposition – Invertibility of a diagonal matrix

Let

$$D = \text{diag}(d_1, \dots, d_n).$$

Then D is invertible if and only if $d_i \neq 0$ for every i . In that case,

$$D^{-1} = \text{diag}\left(\frac{1}{d_1}, \dots, \frac{1}{d_n}\right).$$

Proof. Suppose first that every $d_i \neq 0$, and define

$$B = \text{diag}\left(\frac{1}{d_1}, \dots, \frac{1}{d_n}\right).$$

The product of two diagonal matrices is diagonal, with diagonal entries equal to the products of corresponding diagonal entries. Hence

$$DB = BD = I_n.$$

Thus D is invertible and $B = D^{-1}$.

Conversely, suppose $d_j = 0$ for some j . Let e_j be the j th standard basis vector. Then $e_j \neq 0$ but

$$De_j = 0.$$

Thus $\ker(D) \neq \{0\}$, so D is not injective and therefore not invertible. □

6 Invertible matrices

6.1 Uniqueness of the inverse

Proposition – The inverse is unique
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If $A \in \mathbb{R}^{n \times n}$ is invertible, then its inverse is unique.

Proof. Suppose B and C are both inverses of A . Then

$$AB = BA = I_n, \quad AC = CA = I_n.$$

Using associativity,

$$B = BI_n = B(AC) = (BA)C = I_n C = C.$$

Therefore the inverse is unique. □

6.2 Algebraic properties of inverses

Proposition – Properties of invertible matrices

Let $A, B \in \mathbb{R}^{n \times n}$ be invertible, and let $\alpha \in \mathbb{R} \setminus \{0\}$. Then

- (1) $(A^{-1})^{-1} = A$;
- (2) $(AB)^{-1} = B^{-1}A^{-1}$;
- (3) $(A^\top)^{-1} = (A^{-1})^\top$;
- (4) $(\alpha A)^{-1} = \frac{1}{\alpha}A^{-1}$;
- (5) $I_n^{-1} = I_n$.

Proof.

1. *Inverse of the inverse.* Since

$$A^{-1}A = AA^{-1} = I_n,$$

the matrix A is an inverse of A^{-1} . By uniqueness of the inverse, $(A^{-1})^{-1} = A$.

2. *Inverse of a product.* We verify both products:

$$\begin{aligned} (AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} = AI_nA^{-1} = I_n, \\ (B^{-1}A^{-1})(AB) &= B^{-1}(A^{-1}A)B = B^{-1}I_nB = I_n. \end{aligned}$$

Thus $(AB)^{-1} = B^{-1}A^{-1}$.

3. *Inverse of a transpose.* Using $(XY)^\top = Y^\top X^\top$,

$$(A^{-1})^\top A^\top = (AA^{-1})^\top = I_n,$$

and

$$A^\top (A^{-1})^\top = (A^{-1}A)^\top = I_n.$$

Hence $(A^\top)^{-1} = (A^{-1})^\top$.

4. *Inverse of a nonzero scalar multiple.* Since $\alpha \neq 0$,

$$(\alpha A) \left(\frac{1}{\alpha} A^{-1} \right) = AA^{-1} = I_n,$$

and similarly in the reverse order. Therefore

$$(\alpha A)^{-1} = \frac{1}{\alpha} A^{-1}.$$

5. *Identity matrix.* Since $I_n I_n = I_n$, the identity matrix is its own inverse. □

6.3 Equivalent characterizations of invertibility

Theorem – Invertible matrix theorem

Let $A \in \mathbb{R}^{n \times n}$. The following statements are equivalent:

- (1) A is invertible;
- (2) $\ker(A) = \{0\}$;
- (3) $\text{rank}(A) = n$;
- (4) the columns of A are linearly independent;
- (5) the rows of A are linearly independent;
- (6) the columns of A span $\mathbb{R}^n$, i.e. $\text{range}(A) = \mathbb{R}^n$;
- (7) $x \mapsto Ax$ is bijective;
- (8) $x \mapsto Ax$ is surjective;
- (9) for every $b \in \mathbb{R}^n$, the equation $Ax = b$ has a unique solution;
- (10) A is row-equivalent to I_n .

Proof. We organize the proof around the full-rank condition $\text{rank}(A) = n$.

Kernel, rank and injectivity. By rank-nullity,

$$n = \text{rank}(A) + \dim(\ker(A)).$$

Hence

$$\text{rank}(A) = n \iff \ker(A) = \{0\}.$$

By the kernel criterion, this is also equivalent to injectivity of $x \mapsto Ax$.

Columns. The range of A is the span of its n columns. Therefore

$$\text{rank}(A) = n \iff \text{the columns span } \mathbb{R}^n.$$

Since there are exactly n columns in the n -dimensional space $\mathbb{R}^n$, this is also equivalent to the columns being linearly independent.

Rows. Row rank equals column rank. Hence

$$\text{rank}(A) = n \iff \dim(\text{Row}(A)) = n.$$

Because A has exactly n rows, this is equivalent to the rows being linearly independent.

Surjectivity and bijectivity. The map $x \mapsto Ax$ is surjective if and only if $\text{range}(A) = \mathbb{R}^n$, hence if and only if $\text{rank}(A) = n$. For a linear map $\mathbb{R}^n \rightarrow \mathbb{R}^n$, full rank is therefore equivalent to both injectivity and surjectivity, hence to bijectivity.

Existence and uniqueness of solutions. For a fixed $b \in \mathbb{R}^n$, solving $Ax = b$ means finding a preimage of b under the map $x \mapsto Ax$. Thus “for every b , a solution exists” is exactly surjectivity, and “for every b , that solution is unique” adds injectivity. Therefore statement (9) is equivalent to bijectivity.

Row equivalence to the identity. A square matrix is row-equivalent to I_n if and only if its RREF is I_n . This happens if and only if there is a pivot in every column, i.e. if and only if

$$\text{rank}(A) = n.$$

Invertibility. If A is invertible, then for every $b \in \mathbb{R}^n$,

$$x = A^{-1}b$$

is a solution of $Ax = b$. It is unique: if $Ax = Ay$, multiplication by A^{-1} gives $x = y$. Thus invertibility implies statement (9).

Conversely, assume $x \mapsto Ax$ is bijective. Its inverse map

$$T_A^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

is linear. Let B be its matrix in the standard basis. Since

$$T_A^{-1} \circ T_A = \text{Id}_{\mathbb{R}^n} \quad \text{and} \quad T_A \circ T_A^{-1} = \text{Id}_{\mathbb{R}^n},$$

the matrix-of-a-composition rule gives

$$BA = I_n, \quad AB = I_n.$$

Hence A is invertible and $B = A^{-1}$.

All ten statements are therefore equivalent. □

NB: In lecture 3 about determinants we will prove the following additional equivalent condition for invertibility:

$$\det(A) \neq 0$$

6.4 Gaussian elimination and the inverse

Proposition – Computing an inverse by row reduction

Let $A \in \mathbb{R}^{n \times n}$. Apply the same elementary row operations to the two blocks of the augmented matrix

$$[A \mid I_n].$$

If the left block is reduced to I_n , then the right block becomes A^{-1} . If the left block cannot be reduced to I_n , then A is not invertible.

Proof. A sequence of elementary row operations is equivalent to left-multiplication by an invertible matrix E . Thus

$$[A \mid I_n] \mapsto [EA \mid EI_n] = [EA \mid E].$$

If the left block becomes I_n , then $EA = I_n$. By the invertible matrix theorem, A is invertible, and multiplying $EA = I_n$ on the right by A^{-1} gives

$$E = A^{-1}.$$

Therefore the final augmented matrix is

$$[I_n \mid A^{-1}].$$

Conversely, if A is invertible, then $\text{rank}(A) = n$, so its RREF is I_n . Hence failure to reduce the left block to I_n means $\text{rank}(A) < n$, and the invertible matrix theorem shows that A is not invertible. □

7 Change of basis and similarity

7.1 Change of basis for an endomorphism

Theorem – Change of basis

Let E be an n -dimensional vector space, and let

$$\mathcal{B} = (u_1, \dots, u_n), \quad \mathcal{B}' = (v_1, \dots, v_n)$$

be two bases of E . Let $L : E \rightarrow E$ be linear, and set

$$A = [L]_{\mathcal{B}}, \quad B = [L]_{\mathcal{B}'}$$

Define

$$P = [[v_1]_{\mathcal{B}} \cdots [v_n]_{\mathcal{B}}].$$

Then

$$[x]_{\mathcal{B}} = P[x]_{\mathcal{B}'} \quad \text{for all } x \in E,$$

the matrix P is invertible, and

$$B = P^{-1}AP.$$

Proof.

Coordinate change formula. Let

$$[x]_{\mathcal{B}'} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}.$$

Then

$$x = \alpha_1 v_1 + \cdots + \alpha_n v_n.$$

Taking coordinates in the basis $\mathcal{B}$ gives

$$[x]_{\mathcal{B}} = \alpha_1 [v_1]_{\mathcal{B}} + \cdots + \alpha_n [v_n]_{\mathcal{B}} = P[x]_{\mathcal{B}'}.$$

Invertibility of P . The coordinate map $x \mapsto [x]_{\mathcal{B}}$ and the coordinate map $x \mapsto [x]_{\mathcal{B}'}$ are both bijections from E to $\mathbb{R}^n$. Therefore the map sending $[x]_{\mathcal{B}'}$ to $[x]_{\mathcal{B}}$ is a bijective linear map on $\mathbb{R}^n$. Its matrix is P , so P is invertible. Its inverse sends $\mathcal{B}$ -coordinates to $\mathcal{B}'$ -coordinates.

Matrix of L in the new basis. For every $x \in E$,

$$[L(x)]_{\mathcal{B}} = A[x]_{\mathcal{B}}.$$

Using the coordinate change formula for both x and $L(x)$,

$$P[L(x)]_{\mathcal{B}'} = AP[x]_{\mathcal{B}'}$$

Multiplying by P^{-1} on the left gives

$$[L(x)]_{\mathcal{B}'} = P^{-1}AP[x]_{\mathcal{B}'}$$

Since this holds for every x , the matrix of L in $\mathcal{B}'$ is

$$B = P^{-1}AP.$$

□

7.2 General change of bases in domain and codomain

Proposition – General change-of-basis formula

Let $L : V \rightarrow W$ be linear. Let $\mathcal{B}, \mathcal{B}'$ be bases of V and $\mathcal{C}, \mathcal{C}'$ bases of W . Assume

$$[x]_{\mathcal{B}} = P[x]_{\mathcal{B}'}, \quad [y]_{\mathcal{C}} = Q[y]_{\mathcal{C}'}$$

If

$$A = [L]_{\mathcal{C} \leftarrow \mathcal{B}}, \quad A' = [L]_{\mathcal{C}' \leftarrow \mathcal{B}'},$$

then

$$A' = Q^{-1}AP.$$

Proof. For every $x \in V$,

$$[L(x)]_{\mathcal{C}} = A[x]_{\mathcal{B}}.$$

By the two coordinate-change formulas,

$$Q[L(x)]_{\mathcal{C}'} = AP[x]_{\mathcal{B}'}$$

Multiplying by Q^{-1} gives

$$[L(x)]_{\mathcal{C}'} = Q^{-1}AP[x]_{\mathcal{B}'}$$

Therefore

$$A' = Q^{-1}AP.$$

□

7.3 Similar and equivalent matrices

Corollary – Similar matrices represent the same endomorphism in different bases

If A and B are matrices of the same endomorphism $L : E \rightarrow E$ in two different bases, then there exists an invertible matrix P such that

$$B = P^{-1}AP.$$

Thus A and B are similar.

Proof. This is exactly the change-of-basis theorem above, because the same basis change is used in the domain and in the codomain of the endomorphism. □

Remark – Equivalent versus similar matrices

For a general linear map $L : V \rightarrow W$, changing the basis of the domain and the basis of the codomain independently gives

$$A' = Q^{-1}AP,$$

with P and Q invertible. This is the origin of the notion of “**equivalent**” matrices. For an endomorphism $L : E \rightarrow E$, using one basis change in both domain and codomain forces the two change-of-basis matrices to be linked and gives the more restrictive relation

$$B = P^{-1}AP.$$

“**Similarity**” refers to: different matrix representations of the same endomorphism.